

Fractional feedback square, perspective domination, and near Cartan ownership boundary

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

first issued 2026-07-27 · this version issued 2026-07-27 · v1.0

1 Purpose and scope

R-094 paid the value-heat control prefix against a declared fraction θ of the rootwise future-feedback squares. It explicitly left open the identity embedding those rootwise squares into the global R-086 endpoint square. This note resolves that algebraic question and audits the resulting matrix perspective.

There are four exact conclusions. First, a moving-prefix defect separates the sum of future-feedback Doob squares from the global terminal derivative square. The defect contains a low projection, present-prefix squares, and present-future crosses, and has no sign. Second, after only the fraction $1 - \theta$ of the endpoint coefficient square is retained, the R-092 perspective energy is positive if and only if a new matrix domination $2R \succeq \theta B$ holds. Third, the same loss appears after conditioning as an explicit positive resolvent-gap debt. Fourth, the currently proved absolute root estimates admit no scale-dependent fraction that simultaneously pays the R-094 prefix and suppresses a root covariance debt of scale 2^j .

The analysis also fixes the ownership of the surviving R-086 term $T_G^>$. It has an exact divergence plus bounded exterior-curvature homotopy. Its near-root projector is disjoint from the regular Cartan FAR theorem of R-092, so that theorem does not pay $T_G^>$. Conversely, adding an unprojected Cartan square while retaining $T_G^>$ would duplicate the same Taylor/Gram channel.

These are an exact ownership theorem, a restricted positive criterion, and method no-gos. They are not a counterexample to the complete production action. The full rootwise packet with its moving-prefix defect, Wick trace, backward heat, low and paid terms, R-063 forest, and conditional means remains to be bounded. Complete H_N , REG, OVERLAP_{src}, Nelson, a measure, and Sector A remain open. Tier stays T4.

The binding authorities are R-079 for the exact current Doob decomposition, R-086 for the translated-Wick coefficient-dominant packet, R-092 for the matrix-perspective telescope and regular Cartan FAR theorem, R-093 for the augmented conditional normal form, and R-094 for the rootwise fractional square allocation.

2 The exact rootwise packet remains the starting point

Use the R-094 notation

$$i_j = u_j + y_j, \quad \delta_j^D \mathcal{J}_j = p_j + x_j, \quad \tau_j = \frac{1}{2} \mathbb{E} \int (\mathcal{B}_* - \mathcal{B}_j) : \Delta \Gamma_j. \quad (2.1)$$

For $0 < \theta < 1$, define

$$\begin{aligned}\mathcal{N}_{j,\theta} &:= \mathbb{E}\langle p_j + x_j, u_j \rangle + \frac{1}{2}\mathbb{E}\|u_j\|^2 - \tau_j \\ &\quad + \mathbb{E}\langle p_j + u_j, y_j \rangle + \frac{1-\theta}{2}\mathbb{E}\|y_j\|^2, \\ \mathcal{V}_{j,\theta} &:= \mathbb{E}\langle x_j, y_j \rangle + \frac{\theta}{2}\mathbb{E}\|y_j\|^2.\end{aligned}\tag{2.2}$$

R-094 (6.4) is exactly

$$\mathfrak{J}_j = \mathcal{N}_{j,\theta} + \mathcal{V}_{j,\theta}.\tag{2.3}$$

Therefore R-079 (4.4), before any spatial rational projection, is

$$\boxed{\mathbb{E}\mathfrak{B}_{J,\text{safe}}^> = \mathfrak{L}_\ell + \sum_j \{\mathfrak{F}_j + \mathcal{N}_{j,\theta} + \mathcal{V}_{j,\theta}\} - \mathbb{E}\Delta\mathcal{R}_\leq}.\tag{2.4}$$

This is the strongest unconditional once-only embedding presently available. The combined value innovation and backward-heat compensator is owned by p_j, x_j ; the covariance traces are owned by $\mathfrak{F}_j, \tau_j$; and the R-063 lower-chaos forest is internal to the reconstruction of the Wick coefficient. None is an additional correction to append after (2.4).

R-094 proves

$$\sum_j \mathcal{V}_{j,\theta} \geq -\frac{1}{2} \sum_j \theta^{-1} \mathbb{E}\|x_j\|^2, \quad \sum_j \mathbb{E}\|x_j\|^2 \leq C \frac{16}{7} 2^{-3j_0} X.\tag{2.5}$$

What remained conditional was the replacement of the rootwise reserve $\sum_j \|y_j\|^2$ by a fraction of the global R-086 terminal square. The next section computes the exact difference.

3 The moving-prefix square defect

Let the terminal coefficient be held fixed in the following Hilbert-current coordinates:

$$\begin{aligned}\rho_* &= Q_\Pi^{1/2} M_*^T D(A^* - A^0), \\ \rho_j^- &= Q_\Pi^{1/2} M_*^T D(A^{(j)} - A^0), \\ \rho_j^+ &= Q_\Pi^{1/2} M_*^T D(A^* - A^{(j)}).\end{aligned}\tag{3.1}$$

Thus $\rho_* = \rho_j^- + \rho_j^+$ for every j . Put

$$\zeta_j = \delta_j^D \rho_j^-, \quad y_j = \delta_j^D \rho_j^+.\tag{3.2}$$

The symbol y_j agrees with the future derivative-feedback coordinate in the fixed-terminal lift; it is not a spatial Gaussian derivative.

Theorem 3.1 (exact moving-prefix square defect)

For every square-integrable terminal current and every $0 < \theta < 1$,

$$\boxed{\begin{aligned}&\frac{\theta}{2} \mathbb{E}\|\rho_*\|^2 - \frac{\theta}{2} \sum_j \mathbb{E}\|y_j\|^2 \\ &= \frac{\theta}{2} \mathbb{E}\|P_\ell \rho_*\|^2 + \frac{\theta}{2} \sum_j \mathbb{E}\|\zeta_j\|^2 + \theta \sum_j \mathbb{E}\langle \zeta_j, y_j \rangle.\end{aligned}}\tag{3.3}$$

Proof. Applying δ_j^D to $\rho_* = \rho_j^- + \rho_j^+$ gives

$$\delta_j^D \rho_* = \zeta_j + y_j. \quad (3.4)$$

Doob Pythagoras for the fixed terminal random variable ρ_* gives

$$\mathbb{E}\|\rho_*\|^2 = \mathbb{E}\|P_\ell \rho_*\|^2 + \sum_j \mathbb{E}\|\zeta_j + y_j\|^2. \quad (3.5)$$

Expand (3.5), subtract $\sum_j \mathbb{E}\|y_j\|^2$, and multiply by $\theta/2$. □

The right side of (3.3) is not nonnegative. On one centered scalar root ξ with $\mathbb{E}\xi^2 = 1$, take

$$\rho_j^- = \xi, \quad \rho_j^+ = -\xi, \quad \rho_* = 0. \quad (3.6)$$

Then $\zeta_j = \xi$, $y_j = -\xi$, and $P_\ell \rho_* = 0$. At $\theta = 1/2$, both sides of (3.3) equal

$$-\frac{1}{4} = \frac{1}{4} - \frac{1}{2}. \quad (3.7)$$

This is an algebraic method fixture, not a production counterexample. It proves that using $\theta \sum_j \|y_j\|^2/2$ is not the same operation as removing $\theta \|\rho_*\|^2/2$ from the global terminal square.

The substitution is valid in the restrictive pure-future subcase

$$P_\ell \rho_* = 0, \quad \zeta_j = 0 \quad \text{for every } j, \quad (3.8)$$

because then (3.3) vanishes. The full terminal feedback does not satisfy (3.8): its terminal coefficient and later selectors can depend on earlier roots, and the moving prefix is generally nonzero.

4 Fractional matrix perspective and its sharp boundary

Let $B = B^T \succeq 0$, let $R = R^T \succ 0$, and put

$$A_\theta = (1 - \theta)B + 2R, \quad \Theta_{\theta,R}(B) = B - BA_\theta^{-1}B. \quad (4.1)$$

The baseline-restored reduced endpoint is

$$\mathcal{E}_{\theta,R} = \frac{1}{2}G^T B G + G^T B c + \frac{1}{2}c^T A_\theta c - \frac{1}{2}B : \Gamma. \quad (4.2)$$

Theorem 4.1 (exact fractional perspective criterion)

One has

$$\boxed{\mathcal{E}_{\theta,R} = \frac{1}{2}\|c + A_\theta^{-1}BG\|_{A_\theta}^2 + \frac{1}{2}G^T \Theta_{\theta,R}(B)G - \frac{1}{2}B : \Gamma,} \quad (4.3)$$

and

$$\boxed{\Theta_{\theta,R}(B) \succeq 0 \iff 2R \succeq \theta B.} \quad (4.4)$$

Proof. Expansion of the square in (4.3) proves the first identity. Since $A_\theta \succ 0$, the Schur complement gives

$$\Theta_{\theta,R}(B) \succeq 0 \iff \begin{pmatrix} B & B \\ B & A_\theta \end{pmatrix} \succeq 0. \quad (4.5)$$

The quadratic form of the block matrix is

$$(x + y)^T B(x + y) + y^T (2R - \theta B)y, \quad (4.6)$$

so (4.5) is equivalent to $2R - \theta B \succeq 0$. □

The failure below the threshold is already exact in one dimension. Let

$$B = 1, \quad \theta = \frac{1}{2}, \quad R = \frac{1}{8}, \quad G = 1, \quad \Gamma = 0. \quad (4.7)$$

Then

$$A_\theta = \frac{3}{4}, \quad \Theta_{\theta,R} = -\frac{1}{3}, \quad c_* = -\frac{4}{3}, \quad (4.8)$$

and the optimized value of (4.2) is $-1/6$. The threshold is exactly $R = 1/4$.

The new domination cannot be supplied by one fixed matrix in the production problem. On a pure-doublet ray,

$$g(te_1) = t \left(1 - \frac{5}{9} \frac{t^2}{t^2 + e} \right) e_1, \quad B_R(te_1) = 4c_1 g(te_1) \otimes g(te_1), \quad (4.9)$$

and therefore

$$t^{-2} B_R(te_1) \longrightarrow \frac{64c_1}{81} e_1 \otimes e_1. \quad (4.10)$$

No fixed past payment R dominates θB_R for all fields. Choosing R from the same-root B_R breaks the past-measurability used in the R-092 perspective telescope and creates the moving- R_k defect already isolated there.

Corollary 4.2 (restricted positive branch)

The R-092 one-reveal augmented argument survives fractional square use if the derivative displacement is past-measurable, the conditional value law is moment matched, and $2R \succeq \theta B$ almost surely. In that case repeat the R-092 completion with A_θ and $\Theta_{\theta,R}$. This covers the frozen or explicitly coefficient-conditioned branch satisfying the new domination. It does not cover the unbounded multistep production feedback.

5 The conditional mean shadow

Condition on a past sigma field $\mathcal{F}$. In the R-093 notation let

$$\bar{B} = \mathbb{E}_{\mathcal{F}} B, \quad q = \mathbb{E}_{\mathcal{F}}(Bz), \quad \bar{A} = \bar{B} + 2R, \quad \bar{A}_\theta = (1 - \theta)\bar{B} + 2R, \quad (5.1)$$

and let

$$r = \mathbb{E}_{\mathcal{F}} \Theta_R(B)^{1/2} z, \quad \Theta_R(B) = B - B(B + 2R)^{-1} B. \quad (5.2)$$

Define the reduced conditional raw block

$$\mathcal{S}_\theta(c) = \frac{1}{2} \{ \mathbb{E}_{\mathcal{F}} z^T B z - \bar{B} : \Gamma \} + c^T q + \frac{1}{2} c^T \bar{A}_\theta c. \quad (5.3)$$

Theorem 5.1 (conditional resolvent-gap identity)

If $\mathfrak{D}_R$ denotes the complete R-093 augmented density, then

$$\begin{aligned} \mathcal{S}_\theta(c) = & \frac{1}{2} \mathfrak{D}_R + \frac{1}{2} |r|^2 + \frac{1}{2} \|c + \bar{A}_\theta^{-1} q\|_{\bar{A}_\theta}^2 \\ & - \frac{1}{2} q^T (\bar{A}_\theta^{-1} - \bar{A}^{-1}) q. \end{aligned} \quad (5.4)$$

Moreover,

$$\begin{aligned}
0 \leq G_\theta(q) &:= \frac{1}{2} q^T (\bar{A}_\theta^{-1} - \bar{A}^{-1}) q \\
&= \frac{1}{2} \int_{1-\theta}^1 \|\bar{B}^{1/2} (s\bar{B} + 2R)^{-1} q\|^2 ds \\
&\leq \frac{\theta}{2(1-\theta)} q^T \bar{A}^{-1} q.
\end{aligned} \tag{5.5}$$

Proof. Complete the c square in (5.3), then add and subtract the two mean debts in R-093 (2.5). For (5.5), differentiate $q^T (s\bar{B} + 2R)^{-1} q$ in s and integrate. Finally $\bar{A}_\theta \succeq (1-\theta)\bar{A}$ gives the last bound. $\square$

Thus the two R-093 mean debts are not missing-baseline artifacts. With the full square, their exact endpoint companions are visible. Fractional square use creates the additional nonnegative debt G_θ . The pointwise loss in (4.3) and the conditional loss in (5.4) are the same resolvent mechanism.

The terminal $|r|^2$ companion is also indispensable. On four equiprobable atoms let

$$\frac{z}{t} = (1, -1, 2, -2), \quad B = (0, 2, 3/2, 1/2), \quad R = 1. \tag{5.6}$$

Then

$$\bar{B} = 1, \quad \Gamma = \frac{5}{2} t^2, \quad q = 0, \quad \mathbb{E}(Bz^2) = \bar{B}\Gamma, \tag{5.7}$$

but

$$r = \frac{t}{4} \left[-1 + 2\sqrt{\frac{6}{7}} - 2\sqrt{\frac{2}{5}} \right] \neq 0. \tag{5.8}$$

Hence $\mathfrak{D}_R = -|r|^2$, while restoring the terminal mean energy makes the complete raw Wick contribution zero.

For terminal future feedback, later selectors may depend on root j . Taking the past at $j-1$ retains $\Delta\Gamma_j$ but the derivative displacement is not past-measurable; taking it at j restores measurability but loses that covariance increment. This filtration mismatch is exactly why the one-reveal positive branch does not prove multistep H_N .

6 Scale-dependent fractions are critical under the present bounds

Let $\theta_j = \theta_* 2^{-\alpha(j-j_0)}$. R-094 proves only

$$\mathbb{E}\|x_j\|^2 \lesssim 2^{-j} \mathbb{E}\|H_j\|_2^2, \quad H_j = \sum_{k \leq j} Da_k, \tag{6.1}$$

and the shell smoothing gives $\|Da_k\|_2^2 \lesssim 2^{-2k} \|h_k\|^2$. The resulting nonorthogonal triangular Hilbert–Schmidt calculation is

$$\boxed{\sum_j \theta_j^{-1} \mathbb{E}\|x_j\|^2 \lesssim \frac{\theta_*^{-1} 2^{-3j_0}}{(1 - 2^{-(1-\alpha)})(1 - 2^{-(3-\alpha)})} X,} \tag{6.2}$$

and is cutoff-uniform only for $\alpha < 1$.

Conversely, an absolute mean/covariance envelope at the natural derivative root scale 2^j would require

$$\sum_j \theta_j 2^j < \infty, \tag{6.3}$$

which for this schedule requires $\alpha > 1$. At $\alpha = 1$ both summands are constant in j . More invariantly, the two per-root weights

$$\frac{2^{-j}}{\theta_j}, \quad \theta_j 2^j \tag{6.4}$$

have product one, so no positive sequence makes both corresponding infinite sums finite.

This is a boundary of the present absolute estimates, not a no-go for a joint signed packet. The tempting window $1 < \alpha < 3$ would require the unproved root-local replacement $\|H_j\|^2 \lesssim 2^{-2j}X$ and would delete the old prefixes $k \ll j$. Paying (5.5) from the terminal perspective submartingale would instead reuse the R-092 endpoint already partitioned into its positive increments unless an independent terminal reserve were first proved.

7 Near $T_G^>$ has an exact Cartan homotopy but no separate payment

Let

$$F(z) = q(z)z, \quad q(z) = \frac{z^T S z}{|z|^2 + e}, \quad g(z) = P_{\text{dbl}} z - \frac{5}{9} F(z). \quad (7.1)$$

Put $\varrho = |z|^2 + e$ and $r_S = (S - qI)z$. The exterior curvature of the one-form $F(z) \cdot dz$ is

$$\Omega_z(h, k) = DF(z)[h] \cdot k - DF(z)[k] \cdot h = \frac{2}{\varrho} \{ (r_S \cdot h)(z \cdot k) - (r_S \cdot k)(z \cdot h) \}. \quad (7.2)$$

Since $r_S \wedge z = Sz \wedge z$ and $\|S\|_{\text{op}} \leq 1$,

$$\|\Omega_z\| = \frac{2|r_S \wedge z|}{|z|^2 + e} \leq \frac{2\|S\|_{\text{op}}|z|^2}{|z|^2 + e} \leq 2. \quad (7.3)$$

For $u_t = z + ta$, $G_i = D_i z$, and $c_i = D_i a$, the homotopy formula gives

$$\begin{aligned} & g(z+a) \cdot (G_i + c_i) - g(z) \cdot G_i \\ &= D_i \Psi(z, a) - \frac{5}{9} \int_0^1 \Omega_{u_t}(a, G_i + tc_i) dt. \end{aligned} \quad (7.4)$$

where

$$\Psi(z, a) = \frac{1}{2} \{ (z+a)^T P_{\text{dbl}}(z+a) - z^T P_{\text{dbl}} z \} - \frac{5}{9} \int_0^1 F(u_t) \cdot a dt. \quad (7.5)$$

Thus a divergence shortcut alone is false: the bounded Cartan curvature is load-bearing.

The rational Gram satisfies

$$D^2 B_R[h, h] = 4c_1 \{ D^2 g[h, h] \otimes g + g \otimes D^2 g[h, h] + 2Dg[h] \otimes Dg[h] \}. \quad (7.6)$$

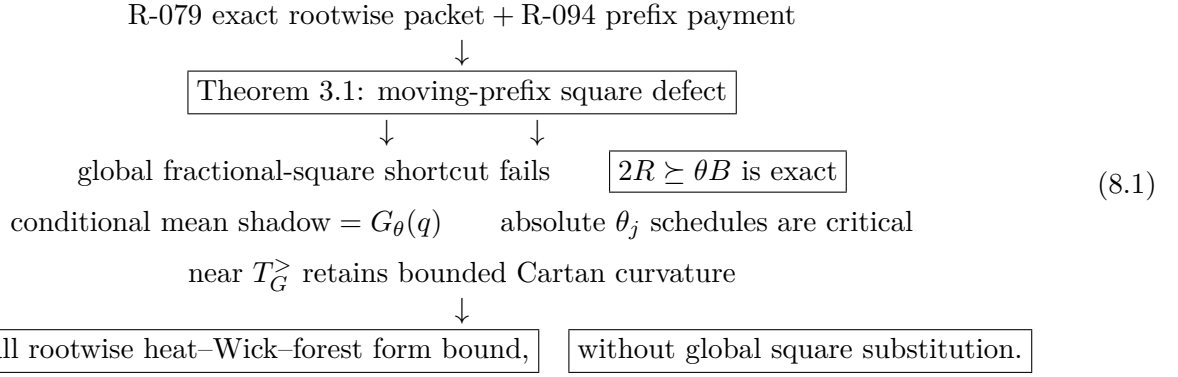
Because $Dg = P_{\text{dbl}} - (5/9)DF$ and $D^2 g = -(5/9)D^2 F$, (7.6) uses the same coefficient primitives as the complete R-087 Cartan mixed variation. The projectors nevertheless keep the proved and open pieces disjoint. R-092 regular Cartan FAR has probability root j and output $m \geq j + 5$; the surviving R-086 $T_G^>$ is a near-root piece with $|m - j| \leq C$ and explicit payload scale $r < j$. Hence FAR cannot be cited as its payment. An unprojected Cartan square cannot be added while $T_G^>$ is also retained.

Conditional centering merely replaces the coefficient by its same-root Doob innovation:

$$\mathbb{E} \langle \Delta_n Dg_j, (\Delta_m K_t)[H_r] \rangle = \mathbb{E} \langle \Delta_n Dg_j, (\Delta_m \delta_j^D K_t)[H_r] \rangle. \quad (7.7)$$

It does not make the term zero. Gaussian integration by parts at the terminal coefficient differentiates later controls and leaves the accepted one-use budget. Therefore $T_G^>$ must remain inside the complete heat-lifted, moving-prefix, perspective/Wick packet.

8 Updated proof and failure map



The route decisions are:

Route	Verdict	Reason / successor
Future square $\rightarrow$ terminal fraction	failed	Exact defect (3.3) has a negative scalar fixture.
Fractional perspective positivity	conditional	Necessary and sufficient domination is $2R \succeq \theta B$; production B is unbounded.
Scale-dependent θ_j	failed by present bounds	Prefix needs $\alpha < 1$, covariance debt needs $\alpha > 1$.
Free mean-debt cancellation	failed	The q resolvent gap and terminal r companion are exact and load-bearing.
Cartan FAR pays $T_G^>$	failed by ownership	FAR and near-root projectors are disjoint; unprojected reuse would duplicate the channel.
Rootwise complete packet	open / required	Keep the moving-prefix defect, full Wick/heat/forest ownership, and conditional means in one form.

9 Devil's-advocate self-test

- Objection: The low and present-prefix terms in (3.3) are positive, so the defect is harmless. UPHELD as false.** The cross $\sum_j \mathbb{E}\langle \zeta_j, y_j \rangle$ has no sign, and (3.6)–(3.7) makes the total defect exactly negative.
- Objection: A small fixed θ preserves the R-092 perspective positivity. UPHELD as false without a new hypothesis.** The exact condition is $2R \succeq \theta B$. Fixture (4.7)–(4.8) fails at $\theta = 1/2$, and production growth (4.10) defeats every fixed R globally.
- Objection: Letting θ_j decay with the root closes both losses. UPHELD as false for the currently proved absolute ledgers.** Equations (6.2)–(6.4) give incompatible strict inequalities. This does not exclude a signed cancellation inside the complete packet.
- Objection: The R-093 mean debts disappear after the B_0 baseline is restored. UPHELD as false.** R-092 restores that baseline before R-093; (5.4) then derives the additional fractional resolvent debt. Fixture (5.6)–(5.8) shows that the terminal mean energy is also indispensable.
- Objection: Regular Cartan FAR already proves $T_G^>$. UPHELD as an ownership error.** The probability-root/output projectors are disjoint. Formula (7.6) explains the shared primitives but does not identify the two projected pieces.

6. **Objection:** These fixtures disprove the complete H_N lower bound. **DISMISSED.** They omit some or all of the heat, low, paid, trace, and R-063 forest companions. They refute only the listed substitutions and termwise positivity routes.

10 Executable evidence

The primary executable uses exact rational arithmetic, finite Rademacher trees, random positive matrices, the scalar perspective fixture, the four-atom mean fixture, and finite critical-fraction sums. The independent executable rebuilds the finite filtrations by leaf enumeration, uses a noncommuting generalized-eigenvalue fixture, and derives every scalar value independently. The integrated verifier hash-pins R-079, R-086, R-092, R-093, and R-094; validates the note, PDF, manifest, claim, result, negative, exploration, task, roadmap, and honesty surfaces; and fails closed on any downstream promotion.

Run from the repository root:

```
python codes/foundations/
  a13_classii_fractional_feedback_square_perspective_boundary_verify.py
```

11 Result footer

Result ID: A13-CLASSII-FRACTIONAL-FEEDBACK-SQUARE-
PERSPECTIVE-DOMINATION-BOUNDARY

Precise statement: The rootwise/global square difference is the signed moving-prefix defect (3.3). Fractional perspective is PSD iff $2R \geq \theta B$; conditioning gives (5.5). Current absolute θ_j schedules are critical. Near $T_G^>$ retains bounded Cartan curvature outside regular FAR.

Scope: Finite cutoff and positive floor; exact R-079/R-086/R-092/R-093/R-094 conventions and algebraic fixtures.

Dependencies: R-079; R-086; R-092; R-093; R-094.

Evidence grade: ANALYTIC, EXACT, EXECUTED, BOUNDARY.

Reproduction command: python codes/foundations/
a13_classii_fractional_feedback_square_
perspective_boundary_verify.py

Expected output: Primary 246, independent 56, integrated 152, aggregate 454 assertions PASS; exit zero.

Falsification gate: Any identity, fixture, ownership, hash, PDF, count, public-link, or T4-scope assertion fails.

Tier before / after: T4 / T4; no promotion.

No-overclaim statement: Complete rootwise form, H_N , REG, H_A , OVERLAP_src, Nelson, removals, measure, Sector A, and T5--T7 are open.

Next required action: Retain the full rootwise square and prove one signed moving-prefix--near--heat--Wick--forest--mean form; only then assemble progressive/revisit H_A .